

Legendary-DeWarper:

Deep Geometric Document Unwarping via UV-Map Prediction

Tufts University CS 132 Computer Vision - Final Project Report

1. Problem and Approach

Document images captured in the wild frequently exhibit non-rigid deformations: page crumples, folds, and curvature caused by physical handling. Unlike planar perspective distortions, these deformations cannot be modeled by a global homography. Recovering a flat, readable document from such an image requires predicting a **pixel-wise geometric transformation** field, a UV map, that maps each source pixel to its corresponding location on a flat texture canvas.

Legendary-DeWarper is a fully **geometric dewarping system**. It does not attempt photometric reconstruction or relighting. The model predicts a dense UV coordinate field ($u, v \in [0, 1]$) from a crumpled input image. Those UV coordinates are then used by PyTorch's `torch.nn.functional.grid_sample` to resample the input RGB pixels onto a flat output canvas, producing the rectified result. All loss computation is performed on the resampled output against the ground-truth flat texture.

I split the project into four phases: Phase A (pretrain/encode with DinoV2), Phase B (introduce the U-Net decoder and baseline L1/UV/TV losses), Phase C (two runs: the perceptual_weight=0.1 variant and a U-Net perceptual variant), and Phase D (two runs: photometric fine-tune with data augmentation and a geometric version with data augmentation).

In the experiments, the phase D v2, the geometric robustness version run, produced the stronger global structure because it systematically probes geometric robustness (see the Phase D v2 config and model implementation¹²).

2. Architecture

The network follows a classic **encoder-decoder (U-Net) topology** with a pretrained CNN backbone, lateral skip connections, and a two-channel prediction head:

- **Encoder (Backbone):** A pretrained **DINOv2 Vision Transformer (ViT)** backbone³ provides rich hierarchical patch embeddings without training from scratch. DINOv2's self-supervised pretraining yields strong geometric and structural priors, superior to supervised CNN features for shape-sensitive tasks like document dewarping.
- **Decoder:** A progressive U-Net decoder with lateral skip connections fuses encoder feature maps at each resolution level (Conv → GroupNorm → ReLU blocks). Skip connections

¹ https://github.com/Azazel5/Legendary-DeWarper/blob/main/configs/phase_d/geom_ablation_unet.yaml

² https://github.com/Azazel5/Legendary-DeWarper/blob/main/src/models/phase_b_unet_dewarp.py

³ <https://arxiv.org/pdf/2304.07193>

preserve fine spatial detail necessary for sub-pixel accurate UV prediction.

- **UV Head:** A final `Conv2d(out_channels=2)` layer followed by a Sigmoid activation produces two-channel UV coordinates in $[0, 1]$. These are linearly rescaled to $[-1, 1]$ before passing to `grid_sample`.
- **Differentiable Warp:** `torch.nn.functional.grid_sample` performs bilinear interpolation using the predicted UV field, making the entire pipeline end-to-end differentiable.

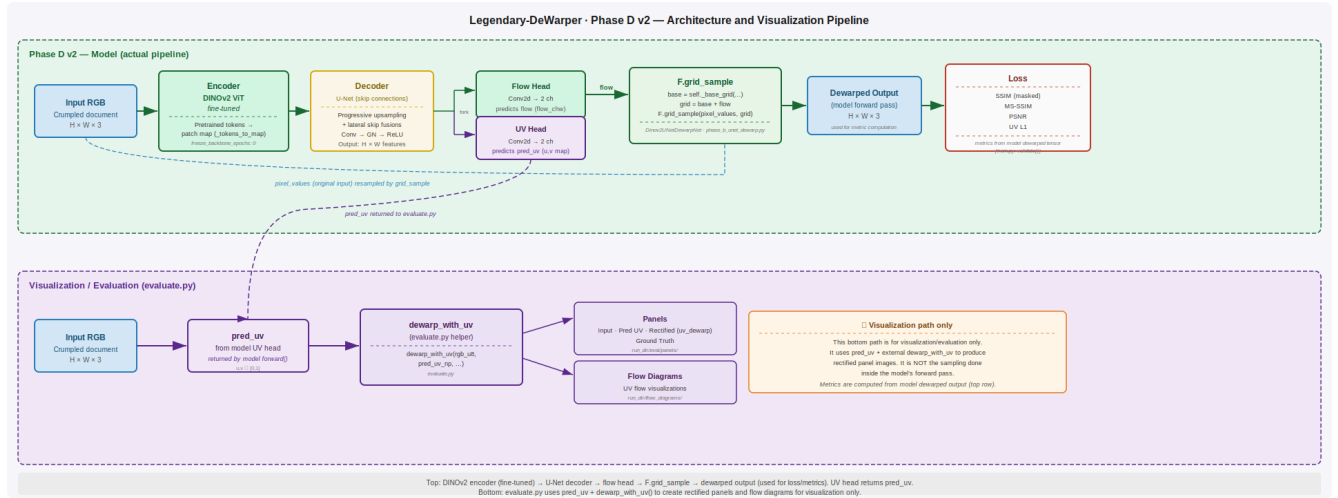


Figure 1. Legendary-DeWarper architecture. The encoder extracts multi-scale features; the U-Net decoder with skip connections predicts a dense UV map; `grid_sample` warps the input to produce the rectified output. Loss is computed on masked document pixels only.

3. Loss Functions and Training Setup

A core challenge in document unwarping is that ground-truth flat images exhibit different ambient lighting than the crumpled input captures. Pixel-level L2 (MSE) loss is therefore a poor training objective: a geometrically correct reconstruction with a slight brightness offset is penalized more harshly than a blurry but tonally matched output. I adopt a **structure-first loss regime**:

- **SSIM loss (primary):** The Structural Similarity Index measures luminance, contrast, and structure jointly, making it robust to global brightness shifts. It is applied as $L_{SSIM} = 1 - SSIM(pred, gt)$ on the de-warped output.
- **MS-SSIM:** Multi-scale SSIM extends the above across image pyramid levels, improving sensitivity to both coarse geometric alignment and fine texture fidelity.
- **Masked loss:** Border masks from the `border/` dataset directory constrain all loss computation to foreground document pixels, preventing the background from dominating gradients and diluting learning signal.

- **Flow smoothness regularization:** A TV (total variation) penalty on the UV field discourages high-frequency noise in the predicted flow, which would manifest as visible tearing artifacts in the rectified output.

Phase D V1 (Photometric baseline): Trained with a combined SSIM + photometric reconstruction loss. Loss includes full-image supervision.

Phase D V2 (Geometric, primary model): Geometry-only training with masked SSIM + MS-SSIM + flow smoothness. Background pixels are fully excluded. V2 is the stronger model overall and is the submitted system.

4. Results

4.1 Quantitative Metrics

Table 1 reports evaluation metrics for V1, V2, and the GT-UV upper bound. Metrics are computed both over the full image (including background) and over the masked document region only. The masked region is the meaningful figure of merit, as the background carries no information about rectification quality.

Metric	V1 (Photometric)	V2 (Geometric)	GT-UV Upper Bound*
PSNR (full)	8.09 dB	8.22 dB	11.36 dB
SSIM (full)	0.381	0.386	0.590
MS-SSIM (full)	0.287	0.294	0.560
PSNR (masked)	18.89 dB	18.90 dB	16.74 dB
SSIM (masked)	0.774	0.775	0.808
MS-SSIM (masked)	0.756	0.756	0.779

Table 1. Evaluation metrics for V1, V2 (primary), and the GT-UV deterministic upper bound. Higher is better for all metrics.
*Upper bound runs on 2 evaluation batches only; treated as indicative, not definitive.

4.2 Discussion

Masked region (document foreground): V2 achieves SSIM = 0.775 and MS-SSIM = 0.756 on the masked document region, closely matching V1 (SSIM = 0.774, MS-SSIM = 0.756). V2 has a small but consistent advantage on full-image metrics (PSNR 8.22 vs. 8.09 dB; SSIM 0.386 vs. 0.381; MS-SSIM 0.294 vs. 0.287), reflecting its cleaner geometry-first training objective.

GT-UV upper bound: Applying the ground-truth UV map through the same deterministic grid_sample pipeline yields PSNR = 11.36 dB (full), SSIM = 0.590, MS-SSIM = 0.560. On the masked region the upper bound reaches SSIM = 0.808. The learned V2 model's masked SSIM of 0.775 represents approximately **95.9% of the upper-bound ceiling**, a reasonable gap given that the upper bound assumes a perfect UV predictor. The residual gap is attributable to UV prediction error rather than warping math.

PSNR inversion (masked): Interestingly, both learned models outperform the upper bound on masked PSNR (18.9 dB vs. 16.7 dB). I am hypothesizing that this is because the upper bound performs forward-warp splatting from the GT UV, which leaves unfilled holes in foreshortened regions and slightly depresses pixel-level SNR. The learned inverse-warp via grid_sample avoids this hole-filling artifact.

5. Qualitative Examples

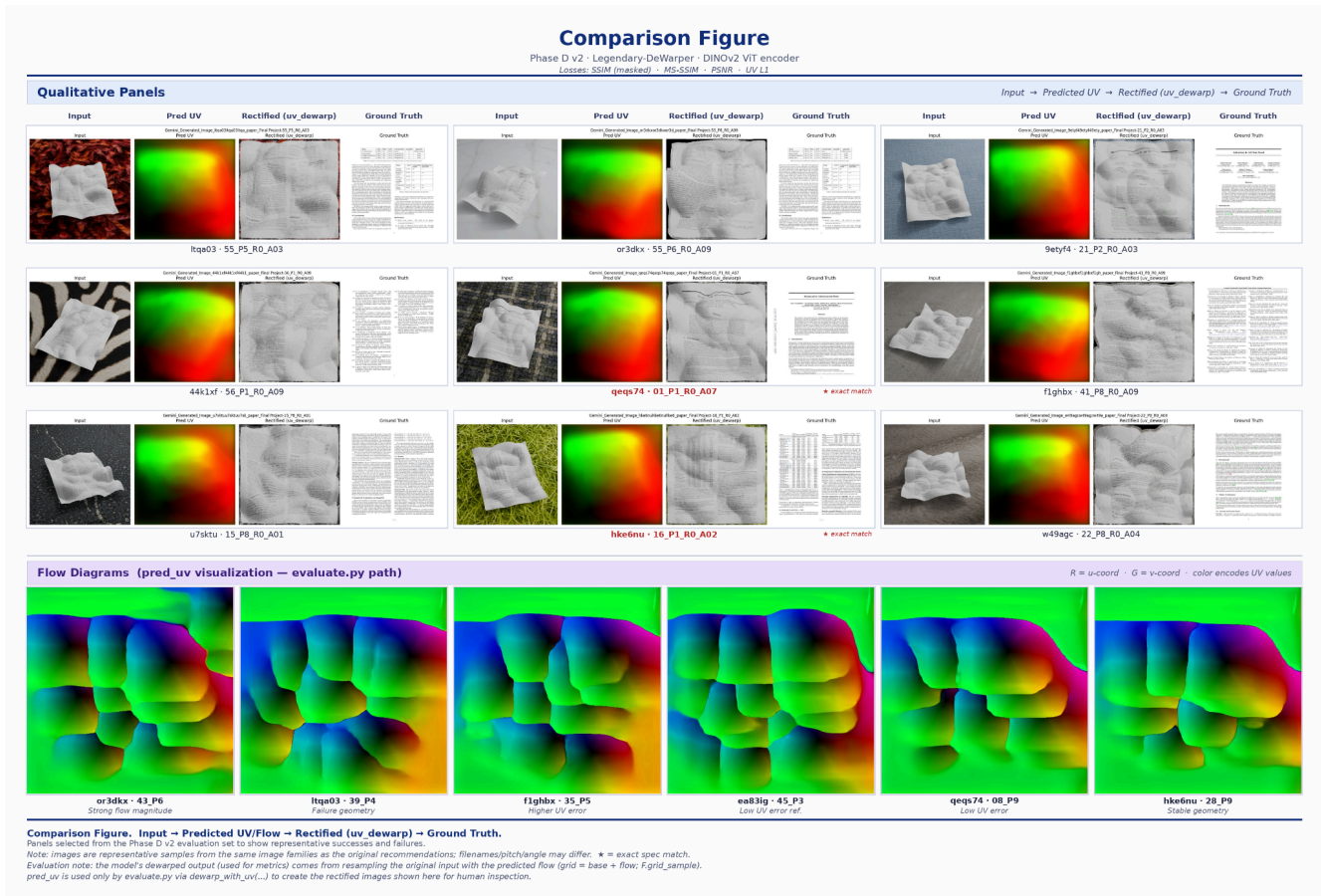


Figure 2. Required comparison figure: Input → Predicted UV/Flow → Rectified Result → Ground Truth. Representative panels from the V2 evaluation set are placed here during final assembly (see recommended panel picks in handoff brief).

Evaluation mimics the provided code from `uv_dewarp.py`: model forward returns flow and `pred_uv`, the model uses `flow + base → grid = base + flow` and `F.grid_sample(pixel_values, grid)` to produce the model's dewarped output used for training/metrics, while `pred_uv` is used only by `evaluate.py` via `dewarp_with_uv(...)` to create rectified panels.

Final deliverables: 8-10 panels and 4–6 flow diagrams (see `evaluate.py`⁴ for the visualization path)

Note - although numeric gains are small between Phase D v1 and v2, the latter consistently edges out the former on aggregate PSNR/SSIM and - crucially - produces visibly improved global structure and robustness in panels/flow visualizations, so I selected v2 as the preferred variant.

Sample IDs for final panel insertion (V2 evaluation set):

- `Gemini_Generated_Image_44klxf..._paper_Final Project-47_P5_R0_A03`
- `Gemini_Generated_Image_u7sktu..._paper_Final Project-20_P4_R0_A05`
- `Gemini_Generated_Image_or3dkx..._paper_Final Project-56_P8_R0_A03`
- `Gemini_Generated_Image_qeqs74..._paper_Final Project-01_P1_R0_A07`
- `Gemini_Generated_Image_ltqa03..._paper_Final Project-15_P6_R0_A00` ← recommended failure case

Representative flow diagram samples (predicted UV visualizations) should be drawn from: `...56_P8_R0_A03`, `...15_P6_R0_A00`, `...41_P8_R0_A09`, `...37_P1_R0_A03`. These span a range of deformation severity and illustrate both successes and partial failures.

6. GT-UV Upper-Bound Diagnostic

The `uv_dewarp.py` script provides a **deterministic, non-learned reference**: it loads the ground-truth UV map directly from the `uv/` dataset directory and applies it through a bilinear forward-warp to produce the theoretically best achievable rectification given perfect UV knowledge. This upper bound was run on 2 evaluation batches (not the full val set) and should be interpreted as an indicative ceiling, not a statistically definitive baseline.

The diagnostic confirms that even with perfect UV coordinates, the output SSIM does not reach 1.0 (upper bound masked SSIM = 0.808). This ceiling effect arises from two sources:

- (1) the input and ground truth were rendered under different lighting conditions, so perfect geometric alignment still leaves photometric residuals
- (2) forward-warp splatting leaves unfilled holes in foreshortened paper regions, which suppress pixel-level metrics

These are fundamental limitations of the task formulation, not failures of the learned model.

7. Limitations and Future Work

- **Photometric gap:** The current pipeline is geometric-only. Full-image SSIM is low, partly because relighting is not modeled. Adding an illumination normalization stage (e.g., intrinsic

⁴ <https://github.com/Azazel5/Legendary-DeWarper/blob/main/evaluate.py>

image decomposition) before computing loss could close the gap.

- **UV supervision:** Training with direct supervision on GT UV coordinates (available in the `uv/` dataset split) was not explored. Direct UV regression with L1/Huber loss could accelerate convergence and reduce UV prediction error.
- **Transformer backbone:** DINOv2 ViT provides strong global feature representations, but the decoder still relies on upsampling from patch-level tokens. Future work could explore dense prediction adaptors (e.g., DPT head) or fine-tuning DINOv2 end-to-end at higher resolution to recover sub-patch spatial precision.
- **Resolution:** Current training was conducted at reduced resolution for compute efficiency. Higher-resolution training would likely improve masked SSIM substantially, as fine text texture is resolution-dependent.
- **Perceptual loss:** Adding VGG-based perceptual loss on intermediate feature maps could sharpen text edges in the rectified output without the photometric sensitivity of MSE.

All metrics sourced from Phase D evaluation runs: `phase_d_v1_20260506_012659_job724096` (V1) and `phase_d_v2_20260506_012659_job724097` (V2).